

Emergence of Category Theory from Dual Sunyata Structures: The SSZK–KSZS Framework

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2025-12-01

Abstract

We propose a dual-structural foundation for the emergence of category theory based on two mutually generative abstract structures: the “Shiki-Soku-Ze-Ku Category” (SSZK-category), representing the structural aspect of phenomena (form-as-emptiness), and the “Ku-Soku-Ze-Shiki Category” (KSZS-category), representing the transcendental coherence prior to structure (emptiness-as-form). We formulate their duality as an intrinsic mechanism responsible for the emergence of objects, morphisms, and compositional coherence. This dual framework suggests that category theory is not a primitive formalism but an emergent phenomenon resulting from a deeper bidirectional generative process analogous to decoherence/coherence interplay. The framework has connections to foundations of mathematics, quantum theoretic decoherence, and interpretative traditions of Sunyata.

A categorical theory on the emergence of truth, dedicated to Nagarjuna for the insight of Sunyata (Emptiness).

1 Introduction

Category theory is widely regarded as a unifying language of mathematics and theoretical physics. However, the question of whether category theory is itself an emergent structure—rather than a primitive foundation—has remained open. Motivated by conceptual parallels with the dialectics of form and emptiness in Madhyamaka philosophy, as well as modern decoherence theory, we develop a formal system in which category theory emerges from a pair of mutually generative structures.

We introduce two abstract categories: the SSZK-category, capturing stabilized structural relations, and the KSZS-category, capturing pre-structural coherence conditions. Their mutual generation gives rise to the structures recognizable as categorical: objects, morphisms, identity, and composition.

2 Preliminaries

We assume familiarity with standard category theory. Throughout, we use $\mathcal{C}$ to denote an emergent category, while SSZK and KSZS denote pre-categorical structures.

3 The SSZK and KSZS Structures

Definition 1 (SSZK-category). *An SSZK-category is an abstract structure S equipped with stabilized relational data representing phenomenological form. It contains proto-objects and proto-morphisms but not yet full coherence laws.*

Definition 2 (KSZS-category). *A KSZS-category is an abstract structure $\mathbf{K}$ representing coherence potentials that constrain and generate possible relational form. It contains proto-coherence data but no stabilized objects.*

Definition 3 (Dual Sunyata Pair). *A dual Sunyata pair is a pair $(\mathbf{S}, \mathbf{K})$ equipped with mutually generative maps*

$$G : \mathbf{K} \rightarrow \mathbf{S}, \quad F : \mathbf{S} \rightarrow \mathbf{K},$$

such that fixed points of the composite

$$\mathbf{K} \xrightarrow{G} \mathbf{S} \xrightarrow{F} \mathbf{K}$$

constitute emergent categorical coherence.

4 Emergence of Categorical Structure

Theorem 1 (Emergence of Objects). *Under the dual Sunyata pairing $(\mathbf{S}, \mathbf{K})$, stabilized proto-objects in $\mathbf{S}$ that are fixed by the composite FG correspond to objects of an emergent category.*

Theorem 2 (Emergence of Morphisms). *Morphisms of the emergent category arise from proto-morphisms in $\mathbf{S}$ whose coherence constraints in $\mathbf{K}$ stabilize under the duality operator FG .*

Theorem 3 (Emergence of Composition). *Compositionality emerges when KSZS-coherence provides associative constraints that SSZK-structure resolves into stable composites.*

5 Diagrammatic Overview

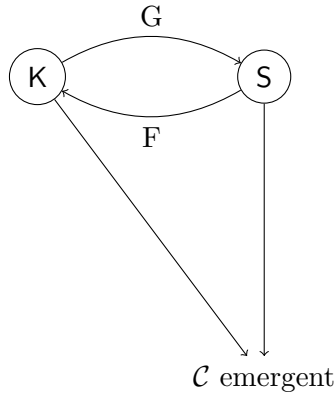


Figure 1: Mutual generation of SSZK and KSZS gives rise to an emergent category.

6 Discussion and Future Work

This framework suggests that categorical structures need not be fundamental. Instead, they arise from dual generative principles reminiscent of decoherence/coherence dynamics. Further work will refine these structures, connect them with topos-theoretic ideas, and explore potential applications to quantum foundations.

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